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	Invoice_ID	Branch	City	Customer_Type
0	SLM-100001	Galle	Galle	Member
1	SLM-100002	Kandy	Kandy	Member
2	SLM-100003	Galle	Galle	Normal
3	SLM-100004	Colombo	Colombo	Member
4	SLM-100005	Kandy	Kandy	Member

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100003 Galle Galle Normal Male Beverages \n", "3 SLM-100004 Colombo Colombo Member Female
Time Payment Rating \n", "0 584.33 5 2629.49 2026-02-15 21:51 Cash 6.8 \n", "1 3170.80 3 9131.90
17 13:40 Cash 8.0 \n", "4 1801.89 6 10378.89 2026-02-16 18:37 eZ Cash 7.9 "] }, "execution_count":
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	Unit_Price	Quantity	Total	Rating
count	1500	1500	1500	1492
mean	1710.407593	3.917333	7126.569253	7.152547
std	1693.475187	1.794244	8727.972507	1.28715
min	121.38	1	137.05	3.1

25%	601.97	2	1841.3675	6.2
50%	1264.445	4	4183.125	7.2
75%	2061.88	5	8921.9925	8
max	9463.72	8	71312.05	10

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Customer_Type			
Member	5189272.02	7477.337205	694
Normal	5500581.86	6824.543251	806

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	Branch	Product_Category	Total
1	Colombo	Electronics	2186796.11
2	Colombo	Food	1355859.09
3	Colombo	Household	973794.46
6	Galle	Electronics	712621.91
8	Galle	Household	528418.86
7	Galle	Food	511215.98
11	Kandy	Electronics	1111450.45
12	Kandy	Food	633688.45
13	Kandy	Household	566660.6

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Household 528418.86\n", "7 Galle Food 511215.98\n", "11 Kandy Electronics 1111450.45\n", "12 Kar
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df.groupby([\"Branch\" , \"Product_Category\"]) [\"Total\"].sum().reset_index()\n", "category_totals = ca
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	Branch	Hour	Total
6	Colombo	14	716945.79
7	Colombo	15	631204.41
9	Colombo	17	607513.35
23	Galle	17	379704.87
24	Galle	18	235803.86
20	Galle	14	195755.45
36	Kandy	16	340638.16
38	Kandy	18	306539.61
39	Kandy	19	302577.37

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Kandy 16 340638.16\n", "38 Kandy 18 306539.61\n", "39 Kandy 19 302577.37"] }, "execution_count"
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in the afternoon and evening.\n", "- Cash dominates as the preferred payment method.\n", "- Custome
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such as Galle.\n", "- Allocate more staff during peak hours to reduce congestion and improve service.
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Gender	Product_Category	Unit_Price	Quantity	Total	Date	Time	Payment
Male	Household	584.33	5	2629.49	15/02/2026	21:51	Cash
Male	Household	3170.8	3	9131.9	14/01/2026	15:55	Cash
Male	Beverages	925.68	6	4998.67	16/02/2026	19:45	Card
Female	Personal Care	2521.37	6	17851.3	17/02/2026	13:40	Cash
Male	Household	1801.89	6	10378.89	16/02/2026	18:37	eZ Cash

0001 Galle Galle Member Male Household \n", "1 SLM-100002 Kandy Kandy Member Male Household \n", "2
Personal Care \n", "4 SLM-100005 Kandy Kandy Member Male Household \n", "\n", " Unit_Price Quantity Tot
2026-01-14 15:55 Cash 7.7 \n", "2 925.68 6 4998.67 2026-02-16 19:45 Card 6.7 \n", "3 2521.37 6 17851.30 2
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Strip leading and trailing whitespace from the \"Product_Category\" column\n", "df[\"Product_Category\"] = df
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/ and sort the results in descending order, then display the top 3 categories for each branch\n", "category_tota
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rforming categories like electronics.\n", "- Improve marketing and operational efficiency in lower-performing br
.\n", "- Enhance digital payment options to encourage non-cash transactions.\n", "- Focus on improving custom
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- Ignition

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